

Pathik Compound Group — Common Protein Target Analysis

Source file: Group_Common_Proteins_All_Thresholds.csv **Method:** For each compound group, DrugBank analogs were identified by Tanimoto similarity (threshold ≥ 0.3 , top-20 per compound). Proteins ranking in the top-k (by descending druggability score) for **all** common DrugBank analogs were intersected to yield group-level target predictions. Analysis was run at three thresholds: top-200, top-500, and top-1000.

Group Overview

Group	Compounds	# Compounds	# Common DrugBank Analogs
1	Compound_2, Compound_3, Compound_31, Compound_53	4	14
2	Compound_5, Compound_6	2	10
3	Compound_8, Compound_4, Compound_7	3	2
4	Compound_13, Compound_19, Compound_28	3	6
5	Novel1, Novel2, Novel3	3	10
6	Novel4, Novel5	2	0

Group 6 produced no common analogs at the Tanimoto threshold and is excluded from downstream analysis.

Common Protein Counts per Group

Group	Top-200	Top-500	Top-1000
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1	24	55	100
2	24	59	118
3	76	193	393
4	25	65	147
5	24	49	97
6	0	0	0

Group 3 consistently yields the largest intersection, likely because it has only 2 common DrugBank analogs (Tariquidar and Pagocloner), making the intersection less stringent.

Known Drug Targets in Top Hits

Proteins appearing at top-200 (most stringent — common to all groups 1-5)

These proteins are shared across nearly every group at the strictest threshold and represent the most robust predictions.

Protein	Full Name	Known Drug Target?	Example Drugs
AR	Androgen Receptor	Yes	Enzalutamide, bicalutamide, apalutamide
TNF	Tumour Necrosis Factor	Yes	Adalimumab, etanercept, infliximab
EGF	Epidermal Growth Factor	Yes (via EGFR)	Cetuximab, erlotinib, gefitinib
F2	Prothrombin (Factor II)	Yes	Dabigatran, argatroban
F3	Tissue Factor (Factor III)	Yes	Factor VIIa inhibitors
F5	Coagulation Factor V	Yes	Anticoagulation pathway target
F7	Coagulation Factor VII	Yes	Eptacog alfa

F8	Coagulation Factor VIII	Yes	Emicizumab, recombinant FVIII
F9	Coagulation Factor IX	Yes	Fitusiran, recombinant FIX
F10	Coagulation Factor X	Yes	Rivaroxaban, apixaban, edoxaban
F11	Coagulation Factor XI	Yes	Abelacimab (in trials)
C3	Complement Component 3	Yes	Pegcetacoplan (FDA approved)
C5	Complement Component 5	Yes	Eculizumab, ravulizumab (FDA approved)
C2, C6, C7, C9	Complement pathway	Partial	C6/C7/C9 targeted experimentally
NF1	Neurofibromin 1	Indirect	Loss activates RAS/MEK — targeted via MEK inhibitors
SI, CS, GC, HP, MB, ALB, TH, CDA, KL, TG	Various	Generally no	Housekeeping / plasma proteins — likely scoring noise

Additional known targets emerging at top-500

Protein	Full Name	Known Drug Target?	Example Drugs / Context
BCL2	B-cell lymphoma 2	Yes	Venetoclax, navitoclax
MYC	MYC proto-oncogene	Emerging	BET inhibitors (indirect); direct inhibitors in trials
AKT1	AKT serine/threonine kinase 1	Yes	Ipatasertib, capivasertib
MET	MET receptor tyrosine kinase	Yes	Crizotinib, cabozantinib,

			tepotinib
ALK	Anaplastic lymphoma kinase	Yes	Crizotinib, alectinib, lorlatinib
HRAS / NRAS	RAS GTPases	Yes	Pathway targeted via MEK/ERK inhibitors
JUN / FOS	AP-1 transcription factors	Indirect	Downstream of many oncogenic pathways
NFKBIA	I κ B α (NF- κ B inhibitor)	Indirect	NF- κ B pathway — bortezomib targets upstream
RB1	Retinoblastoma protein	Indirect (Group 3)	Loss drives CDK4/6 dependence — palbociclib context

Additional known targets emerging at top-1000

Protein	Full Name	Known Drug Target?	Example Drugs / Context
MTOR	mTOR kinase	Yes	Rapamycin, everolimus, temsirolimus
SRC	SRC tyrosine kinase	Yes	Dasatinib, bosutinib
MAPK1 / MAPK3	ERK1/2	Yes	Ulixertinib, MK-8353
STAT3	Signal transducer and activator 3	Emerging	Multiple STAT3 inhibitors in trials
CTNNB1	β -catenin	Emerging	Wnt pathway inhibitors in trials
CDK1 / CDK5	Cyclin-dependent kinases	Yes	CDK inhibitors (palbociclib, ribociclib class)
TP53	Tumour protein p53	Indirect (Group 3)	MDM2 inhibitors restore TP53 activity
IKBKG	NF- κ B essential modulator	Indirect (Group 3)	Upstream of NF- κ B survival signalling

PTEN	Phosphatase and tensin homolog	Indirect	Loss activates PI3K/AKT; targeted via AKT inhibitors
FGFR1	FGF receptor 1	Yes (Group 5)	Erdafitinib, pemigatinib
ESR1	Estrogen receptor 1	Yes (Group 3)	Tamoxifen, fulvestrant, elacestrant
PLK1	Polo-like kinase 1	Yes (Group 3)	Volasertib, onvansertib
KRAS	KRAS GTPase	Yes (Group 3)	Sotorasib, adagrasib

Key Findings by Group

Group 1 (Compound_2, _3, _31, _53)

- Top-200 hits dominated by coagulation factors and complement proteins alongside AR and EGF.
- BCL2 and AKT1 emerge at top-1000, suggesting these compounds may engage apoptosis and survival signalling.
- MYC, JUN, FOS, TNF, REL also appear at top-1000, pointing to broad transcriptional/inflammatory pathway involvement.

Group 2 (Compound_5, Compound_6)

- Similar coagulation/complement profile at top-200 as Group 1.
- At top-1000, BCL2, HRAS, MYC, MET, JUN, FOS, and MAX appear, indicating overlap with oncogenic signalling networks.
- NFKBIA at top-500/1000 suggests NF-κB pathway relevance.

Group 3 (Compound_8, _4, _7)

- Largest intersection at all thresholds (76 / 193 / 393 proteins) due to only 2 DrugBank analogs.
- Most diverse target profile: BCL2 (top-500), RB1, CDK5, AKT1 (top-500), and TP53, IKBKG, CTNNB1, KRAS, NRAS, MAPK1/3, ESR1, PLK1, PTEN, STAT3 at top-1000.
- Suggests these compounds may act on broad oncogenic programmes spanning cell cycle, apoptosis, RAS/MAPK, and Wnt signalling.

Group 4 (Compound_13, _19, _28)

- TNF prominent at top-200 alongside coagulation/complement targets and AR.
- ALK and MET appear at top-500, suggesting receptor tyrosine kinase relevance.
- BCL2 and MTOR emerge at top-1000, consistent with thalidomide/lenalidomide/pomalidomide analogs (immunomodulatory drugs with known MTOR and anti-apoptotic activity).

Group 5 (Novel1, Novel2, Novel3)

- BCL2 appears early — at top-500 — making it the most prominent actionable target for this group.
- MTOR, SRC, MAPK3, MET, FGFR1, AKT1, HRAS also appear at top-1000.
- STAT3 appears at both top-500 and top-1000, consistent with vinca alkaloid-like DrugBank analogs that disrupt mitotic/survival signalling.

Group 6 (Novel4, Novel5)

- No common DrugBank analogs found at Tanimoto threshold ≥ 0.3 . No protein predictions possible. Consider lowering the threshold or expanding the reference database.

Summary Assessment

The predicted top targets are **largely biologically plausible**:

- **Coagulation factors and complement proteins** dominate at top-200 across all groups — these are well-validated drug target families and their consistent appearance suggests the scoring captures known pharmacology.
- **AR and TNF** appear at top-200 in every group — both are major drug target classes, though their relevance may reflect broad scoring rather than group-specific biology.
- **BCL2** correctly emerges in all groups (top-500 or top-1000), consistent with the senolytic focus of the project and the known activity of venetoclax/navitoclax.
- **MTOR, AKT1, SRC, MAPK1/3, MET, FGFR1** — all clinically actionable kinases — appear at top-1000, supporting the druggability framework.
- **TP53, PTEN, RB1** are tumour suppressors and are not directly druggable but their appearance reflects oncogenic context.

- Housekeeping proteins (ALB, SI, GAPDH, CS, HP, MB) appearing consistently are likely artifacts of the scoring methodology rather than true target predictions.